

The Costs of Production

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The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 14** — The Costs of Production.

Read the chapter alongside these slides.

Where we are going

1. **What a firm's costs really are** — the economist counts what the accountant leaves out.
2. **The production function** — workers in, components out, and each worker adds less.
3. **From production to cost** — why the cost of one more unit keeps climbing.
4. **The family of cost measures** — total, fixed, variable, average, marginal.
5. **Marginal pulls average** — the crossing that every exam asks about.
6. **The long run** — when the factory itself is up for grabs.
7. **Pulling it together** — one chain from technology to cost curves.

PART 1

What a Firm's Costs Really Are

Why open the firm up?

The supply curve summarized firm behavior: a higher price, a larger quantity supplied. It never said **why**.

The why lives in the firm's costs. They decide three things:

- whether the **next unit** is worth producing,
- what **price** the firm must fetch to survive,
- how the firm behaves as it **grows**.

Behind every supply decision is a cost schedule.

The goal: profit

Profit. Total revenue minus total cost: $\pi = TR - TC$, where $TR = P \times Q$.

Our maintained assumption: **firms maximize profit.**

Not because owners think of nothing else — but because profit maximization predicts observed firm behavior remarkably well.

Revenue is the easy half. The work ahead is the other half: **what belongs in total cost?**

Two kinds of cost

Explicit costs. Input costs that require a direct money payment: wages, rent, materials, interest on a loan.

Implicit costs. Input costs with no money payment — the value of what the owner gives up: the salary not earned elsewhere, the interest the invested savings could have earned.

An **opportunity cost** is whatever is given up to get something. **The economist counts both kinds. The accountant counts only the explicit.**

The engineer who left

A software engineer leaves a \$150,000 job to consult on her own: office rent, equipment, and \$200,000 of savings that had been earning 5 percent.

	Item	\$/year
Explicit	office rent	30,000
Explicit	equipment	20,000
Implicit	salary given up	150,000
Implicit	interest given up: $0.05 \times 200,000$	10,000
Total opportunity cost		210,000

The accountant sees \$50,000. The economist sees **\$210,000**.

The cost of capital: own it or borrow it

A firm buys \$500,000 of equipment. The interest rate is 5 percent.

- **Borrowed:** the firm pays \$25,000 interest per year — an **explicit** cost.
- **Owner's own money:** no payment leaves — but the money stops earning \$25,000 elsewhere — an **implicit** cost.

Same equipment, same \$25,000 economic cost either way.

Only the **accounting** treatment differs.

Two profits

Accounting profit. Total revenue minus explicit costs.

Economic profit. Total revenue minus **all** opportunity costs, explicit and implicit.

The consulting firm earns \$280,000 in revenue:

	Accounting	Economic
Revenue	280,000	280,000
Costs counted	50,000	210,000
Profit	230,000	70,000

The gap between them is **exactly the implicit costs**.

What zero economic profit means

A firm with **zero economic profit** covers every opportunity cost — including the owner's forgone salary and a normal return on the invested capital.

The owner is doing **exactly as well** as in the next-best alternative. Nothing is wrong.

- Economic profit > 0 : better than the alternative — others will want in.
- Economic profit < 0 : worse than the alternative — over time, the owner exits.

Economic profit, not accounting profit, steers decisions.

Active Learning 1

Nadia leaves a \$70,000 job as a staff baker to open her own bakery. Per year: rent \$24,000; ingredients and utilities \$36,000; a part-time assistant \$20,000. She also puts \$50,000 of savings (previously earning 4 percent) into ovens. Revenue: \$160,000.

1. Accounting profit and economic profit. Show the split of costs.
2. Should she keep the bakery open? **Which profit decides, and why — the why is the marks.**
3. At what revenue would she be exactly indifferent?

Four minutes. Paper and pencil.

PART 2

The Production Function

Short run and long run

Short run. At least one input is fixed. The firm can hire workers at short notice; it cannot build a bigger factory at short notice.

Long run. All inputs are variable — factory size included.

Parts 2–5 live in the **short run**: machines and floor space fixed, workers variable.

Part 6 opens the long run.

Write the equation

A firm assembles electronic components. Machines and floor space are fixed; it chooses only L , the number of workers per shift.

Production function. The relationship between inputs used and output produced: here $Q = f(L)$.

$$Q = 24L - L^2 \quad \text{for } 0 \leq L \leq 12$$

24 is the pace a worker sets with the machines to themselves; $-L^2$ is the crowding penalty that grows as the floor fills. Past $L = 12$, another worker would **lower** output.

Plug the numbers in

Workers L	$24L - L^2$	Output Q
0	$24(0) - 0$	0
1	$24(1) - 1$	23
2	$24(2) - 4$	44
3	$24(3) - 9$	63
4	$24(4) - 16$	80
5	$24(5) - 25$	95
6	$24(6) - 36$	108
8	$24(8) - 64$	128
12	$24(12) - 144$	144

Marginal product: what the next worker adds

Marginal product. The extra output from one more unit of input, other inputs held fixed: $MP = \Delta Q / \Delta L$.

The manager's question: **if I add one more worker, how many more components do I get?**

It is a subtraction on the output table. The first worker: $23 - 0 = 23$. The second: $44 - 23 = 21$.

Now run it all the way down...

The marginal product column

L	Q	MP of that worker
1	23	$23 - 0 = 23$
2	44	$44 - 23 = 21$
3	63	$63 - 44 = 19$
4	80	$80 - 63 = 17$
5	95	$95 - 80 = 15$
6	108	$108 - 95 = 13$
7	119	$119 - 108 = 11$
8	128	$128 - 119 = 9$

...and so on down: the 12th worker adds just **1**.

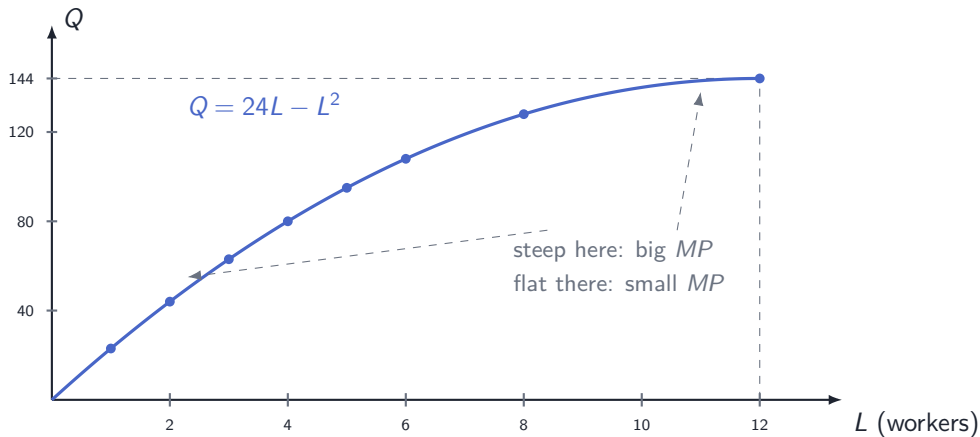
Diminishing marginal product

Diminishing marginal product. As more of a variable input is added, holding other inputs fixed, its marginal product eventually falls.

Why: the **fixed input gets stretched**. Early workers have machines to themselves; later workers share, wait, and coordinate around the same floor.

Every worker adds output — and each adds less than the one before.

Draw the graph



The margin is where the decision lives

The manager never asks “how productive are workers on average?”

The manager asks: **what does one more worker add, and what does one more worker cost?**

- One more worker **adds** MP components.
- One more worker **costs** the wage w .

Next: put a dollar figure on the fixed bill and the wage, and the cost curves build themselves.

Active Learning 2

A print shop's production function: $Q = 30L - 2L^2$ posters per day, for $L = 0$ to 5 workers.

1. **Before computing anything, predict:** will marginal product rise or fall as L grows? One sentence of reasoning.
2. Build the table: Q and MP for $L = 1, \dots, 5$.
3. **Check yourself two ways:** compute Q at $L = 5$ from the formula, then add up your five MP numbers. Why must the two answers agree?

Four minutes. Paper and pencil.

PART 3

From Production to Cost

The firm's two bills

Back to the component assembler. Each shift it pays:

- **\$200** for machines, floor space, insurance — **owed even at zero output**;
- **\$50 per worker** hired.

$$TC = 200 + 50L$$

One bill is stuck; the other moves with the workforce. Costs inherit their shape **from the production function**.

Plug the numbers in

L	Q	Fixed (\$)	Labor $50L$ (\$)	TC (\$)
0	0	200	0	200
1	23	200	50	250
2	44	200	100	300
3	63	200	150	350
4	80	200	200	400
5	95	200	250	450
6	108	200	300	500
8	128	200	400	600
12	144	200	600	800

Rearrange to plot

We want cost **against output** — TC as a function of Q , because output is what the firm sells.

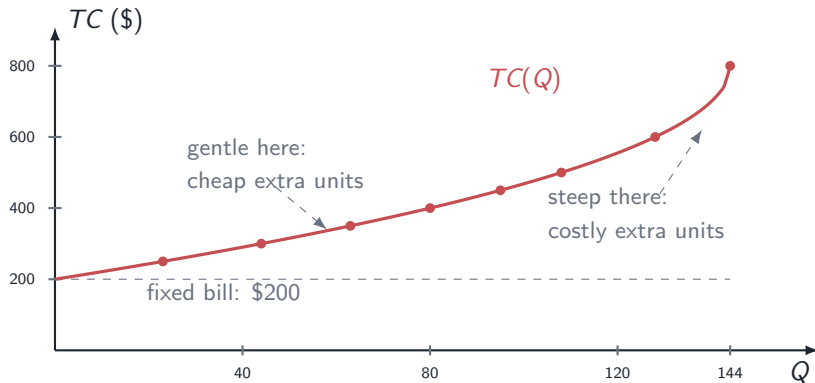
Every row of the table is already a point (Q, TC) :

producing 63 components costs \$350 $\rightarrow (63, 350)$

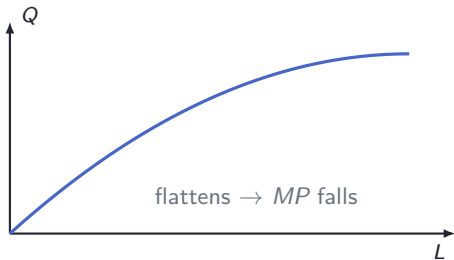
producing 128 components costs \$600 $\rightarrow (128, 600)$

For the smooth curve, the computer finds, for each Q , the workforce that produces it — then charges $200 + 50L$.

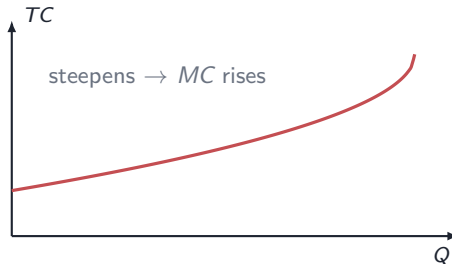
Draw the graph



Mirror image



production: workers $\rightarrow$ output



cost: output $\rightarrow$ dollars

One fact, photographed twice: the production function flattens, so the cost curve steepens.

The cost of one more unit

One more worker costs \$50 and adds MP units — so each of those units costs

$$\text{marginal cost} = \frac{\$50}{MP}$$

Worker	adds (MP)	costs	cost per extra unit
1st	23	\$50	$50/23 = \$2.17$
5th	15	\$50	$50/15 = \$3.33$
7th	11	\$50	$50/11 = \$4.55$
9th	7	\$50	$50/7 = \$7.14$
12th	1	\$50	$50/1 = \$50.00$

MC rises because MP falls

The causal chain, in order:

fixed machines $\rightarrow$ each worker adds less (MP falls) $\rightarrow$ each unit needs more
worker-time $\rightarrow$ **MC rises**

Marginal cost rises for exactly the reason marginal product falls. Same fact, priced.

This is why short-run cost curves eventually climb — for **every** firm with a fixed input.

Active Learning 3

A bagel shop pays every baker \$60 a shift. With a small crew, the marginal product of the next baker is **20 dozen bagels**. With a packed kitchen, the marginal product of the next baker is **4 dozen**.

1. **Predict before you divide:** in which kitchen is the cost of one more dozen higher? One sentence why.
2. Now compute the marginal cost of a dozen in each case.
3. State, in one sentence, the general link between *MP* and *MC*.

Three minutes. Paper and pencil.

PART 4

The Family of Cost Measures

A new firm: the chair workshop

A workshop builds oak chairs. Its daily total cost, in dollars, for Q chairs a day:

$$TC = 100 + 60Q - 12Q^2 + Q^3$$

100 — the daily fixed bill: rent and tool leases, owed at $Q = 0$.

$60Q - 12Q^2 + Q^3$ — the variable bill, shaped so per-unit costs first **fall** (a small crew helps each other) and then **climb** (the bench space runs out).

Same short-run logic — now we need the **per-unit** measures.

Fixed and variable

Fixed costs (FC). Costs that do not vary with output — owed even at $Q = 0$. Here: $FC = 100$.

Variable costs (VC). Costs that vary with output. Here: $VC = 60Q - 12Q^2 + Q^3$.

$$TC = FC + VC$$

Every dollar of total cost is one or the other — **no third category**.

Plug the numbers in

Q	$100 + 60Q - 12Q^2 + Q^3$	$TC (\$)$
0	$100 + 0 - 0 + 0$	100
1	$100 + 60 - 12 + 1$	149
2	$100 + 120 - 48 + 8$	180
3	$100 + 180 - 108 + 27$	199
4	$100 + 240 - 192 + 64$	212
5	$100 + 300 - 300 + 125$	225
6	$100 + 360 - 432 + 216$	244
7	$100 + 420 - 588 + 343$	275
8	$100 + 480 - 768 + 512$	324
9	$100 + 540 - 972 + 729$	397
10	$100 + 600 - 1200 + 1000$	500

The per-unit measures: cost of the typical unit

Average fixed cost. $AFC = FC/Q$ — the fixed bill spread over the units.

Average variable cost. $AVC = VC/Q$ — the variable bill per unit.

Average total cost. $ATC = TC/Q$ — what the **typical** unit costs.

$ATC = AFC + AVC$ — always, because $TC = FC + VC$.

Marginal cost: cost of the next unit

Marginal cost. The rise in total cost from one more unit: $MC = \Delta TC / \Delta Q$.

Straight from the TC column:

the 1st chair	$149 - 100 = \$49$
the 2nd chair	$180 - 149 = \$31$

ATC : what does the typical unit cost?

MC : what would **one more** cost?

Different questions — and they can have very different answers.

The grand table

Q	TC	$AFC = \frac{100}{Q}$	$AVC = \frac{VC}{Q}$	$ATC = \frac{TC}{Q}$	$MC = \Delta TC$
0	100	—	—	—	—
1	149	100.00	49.00	149.00	49
2	180	50.00	40.00	90.00	31
3	199	33.33	33.00	66.33	19
4	212	25.00	28.00	53.00	13
5	225	20.00	25.00	45.00	13
6	244	16.67	24.00	40.67	19
7	275	14.29	25.00	39.29	31
8	324	12.50	28.00	40.50	49
9	397	11.11	33.00	44.11	73
10	500	10.00	40.00	50.00	103

Audit one row. $Q = 6$: $VC = 244 - 100 = 144$, so $AVC = 144/6 = 24$; $ATC = 244/6 = 40.67$;
 $MC = 244 - 225 = 19$.

AFC: the shrinking share of the rent

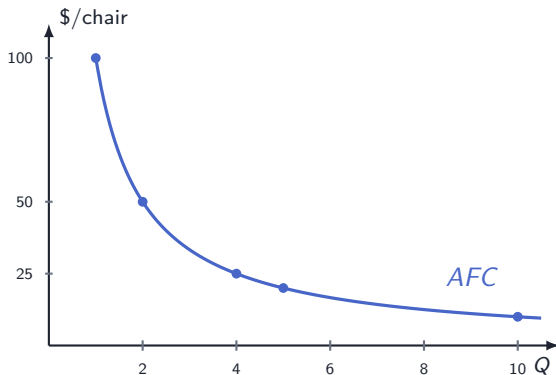
$AFC = 100/Q$ **always falls**: a fixed \$100 spread over ever more chairs.

1 chair: \$100 each

4 chairs: \$25 each

10 chairs: \$10 each

It approaches zero, **never reaches it** — and it has no minimum.



Check the identity

$ATC = AFC + AVC$ — claimed on a definition frame, now **checked in the table**:

Q	AFC	AVC	$AFC + AVC$	ATC
4	25.00	28.00	53.00	53.00 ✓
9	11.11	33.00	44.11	44.11 ✓

Why it must hold: divide $TC = FC + VC$ through by Q .

Habit worth keeping: never trust an identity you have not checked in a row.

What each column answers

Measure	The question it answers	Decision it serves
TC	what does the whole day cost?	the budget
ATC	what does the typical chair cost?	catalog pricing
MC	what would one more cost?	take the next order?

Next part: what happens when the **next**-unit number and the **typical**-unit number meet.

Active Learning 4

Fill in **every** blank. **Write the route to each cell — the route is the marks.**

Deduce FC first; check rows with $ATC = AFC + AVC$.

Q	TC	AFC	AVC	ATC	MC
0	\$60	—	—	—	
1					\$30
2			25		
3				45	
4	170				
5					50

Five minutes. Paper and pencil.

PART 5

Marginal Pulls Average

Watch the turn

ATC falls while *MC* is below it — and turns where *MC* passes it:

<i>Q</i>	<i>MC</i>	<i>ATC</i>	<i>ATC</i> is...
2	31 <	90.00	falling ↓
3	19 <	66.33	falling ↓
4	13 <	53.00	falling ↓
5	13 <	45.00	falling ↓
6	19 <	40.67	falling ↓
7	31 <	39.29	falling ↓ — the bottom
8	49 >	40.50	rising ↑
9	73 >	44.11	rising ↑

Your grade average does the same

Three quizzes in, your average is **80** (240 points). The fourth quiz is the *marginal* quiz:

4th quiz	new average	
60	$(240 + 60)/4 = 75$	below — pulls it down
92	$(240 + 92)/4 = 83$	above — pulls it up
80	$(240 + 80)/4 = 80$	equal — it stands still

The average always moves **toward** the newest mark.

The pulling principle

Each new unit's cost is a *new mark* entering the average.

- $MC < ATC$: the newcomer is cheap — it **pulls the average down**.
- $MC > ATC$: the newcomer is dear — it **pulls the average up**.
- $MC = ATC$: the average **stands still**.

So MC crosses ATC exactly at the minimum of ATC — a consequence, not an assumption.

Check the turn, both directions

The principle predicts; the table must obey. At the bottom, $ATC = 39.29$ at $Q = 7$:

- **Into the bottom:** the 7th chair cost $31 < 40.67$ (the average it joined) — so it had to pull ATC down: $40.67 \rightarrow 39.29$. ✓
- **Out of the bottom:** the 8th chair costs $49 > 39.29$ — so it had to pull ATC up: $39.29 \rightarrow 40.50$. ✓

Two routes to the same turn — prediction and arithmetic agree. If they had not, one of them would be wrong, and we would go looking.

The same pull works on AVC

MC plays the same game against **average variable cost**:

<i>Q</i>	<i>MC</i>	<i>AVC</i>	<i>AVC</i> is...
5	13 <	25.00	falling ↓
6	19 <	24.00	falling ↓ — the bottom
7	31 >	25.00	rising ↑
8	49 >	28.00	rising ↑

MC crosses *AVC* at **its** minimum too — same pull, different average.

But wait: *AVC* bottoms at $Q = 6$, *ATC* at $Q = 7$. **Why not together?**

Why does ATC keep falling after AVC has turned?

Between $Q = 6$ and $Q = 7$, AVC is already rising — yet ATC still falls. Read the pieces off the table:

	ΔAVC	ΔAFC	ΔATC
$Q : 6 \rightarrow 7$	+1.00	-2.38	- 1.38 ↓
$Q : 7 \rightarrow 8$	+3.00	-1.79	+ 1.21 ↑

AFC never stopped falling — it keeps pulling ATC down after AVC has turned. ATC turns only when rising AVC **overpowers** falling AFC .

Two minimums, two outputs — expect to be asked why.

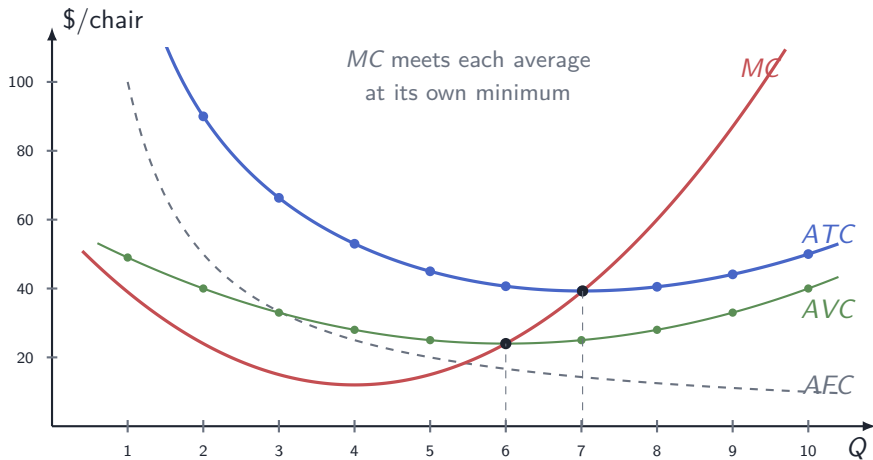
Efficient scale. The quantity at which average total cost is at its minimum.

For the chair workshop: from the table, ATC bottoms at

$$Q = 7 \text{ chairs a day, at } ATC = \$39.29$$

— the output where the workshop makes chairs **as cheaply per chair as it ever can**.
And by the pulling principle, it is where MC meets ATC .

Draw the graph



Three features every firm shares

Strip away this workshop's numbers; three properties survive:

1. *MC* **eventually rises** — diminishing marginal product prices itself into every next unit.
2. *ATC* is **U-shaped** — falling *AFC* dominates early, rising *AVC* dominates late.
3. *MC* **crosses *ATC* at the minimum of *ATC*** — marginal pulls average; the crossing is forced.

These are structural, not accidents of $100 + 60Q - 12Q^2 + Q^3$.

Extensions — the pulling principle by calculus

Everything on this frame is beyond the department's required material. The result was already established from the table; this is the same result, one line of calculus, for those who have met derivatives.

Differentiate $ATC(Q) = TC(Q)/Q$ by the quotient rule:

$$\frac{d(ATC)}{dQ} = \frac{MC \cdot Q - TC}{Q^2} = \frac{MC - ATC}{Q}$$

For $Q > 0$ the sign is the sign of $MC - ATC$: negative when $MC < ATC$, zero when $MC = ATC$, positive when $MC > ATC$ — **the table's pull, in one line.**

Active Learning 5

A consultant studies the chair workshop and reports:

“Marginal cost has been rising since $Q = 5$. Therefore average total cost must be rising too.”

1. Right or wrong? **Find rows of the table that settle it.** The reasoning is the marks — a verdict alone scores nothing.
2. Your quiz average has **fallen** for six weeks running. What do you know for certain about each new quiz score? State the principle that tells you.

Four minutes. Paper and pencil.

PART 6

The Long Run

Fixed today, variable eventually

In the **short run** the factory is given: the only lever is labor, and diminishing MP makes costs climb steeply.

In the **long run** everything is a choice: build a bigger plant, automate, redesign the line. **The fixed bill itself becomes variable.**

Short-run cost curves are drawn *for a given factory*. Change the factory, and the whole curve moves.

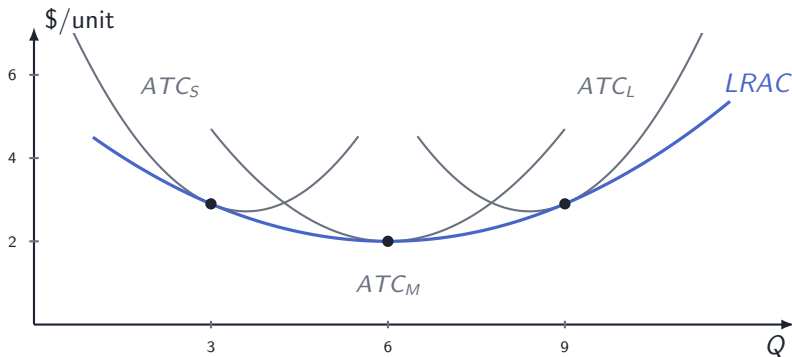
Three factory sizes

A firm can build **small**, **medium**, or **large** — each with its own short-run *ATC* curve.

- **Small**: cheap at low output — overwhelmed at high output.
- **Medium**: best in the middle range.
- **Large**: idle capacity at low output — cheapest at high output.

In the **long run** the firm picks the factory; in the **short run** it is stuck with the one it has.

Draw the envelope



Long-run average cost is the lower envelope: for each Q , the cheapest per-unit cost *any* factory choice can deliver.

Choosing the size

Per-unit cost of each factory, read from the three curves:

Q	Small	Medium	Large
2	\$4.00	\$6.80	\$23.20
6	\$5.60	\$2.00	\$5.60
10	\$23.20	\$6.80	\$4.00

Plan for 2 a day: build **small**. For 6: **medium**. For 10: **large**.

In the long run, the firm rides the envelope.

Economies and diseconomies of scale

Economies of scale. *LRAC falls* as output rises.

Constant returns to scale. *LRAC is flat* as output rises.

Diseconomies of scale. *LRAC rises* as output rises.

On our envelope: economies up to $Q = 6$, diseconomies beyond. Many real industries show a **long flat bottom** between the two.

Why bigger can be cheaper

Specialization. In a large operation, each worker masters one task instead of juggling ten.

Adam Smith opened *The Wealth of Nations* (1776) with a pin factory: ten workers, the pin-making split into distinct operations, producing **tens of thousands of pins a day** — where a lone artisan, doing every step himself, could scarcely make twenty.

Larger scale also justifies **equipment** a small shop could never keep busy.

Why bigger eventually costs more

Coordination. As the organization grows:

- management layers multiply,
- information travels slowly and arrives garbled,
- monitoring effort gets harder — and more expensive.

Eventually coordination costs **outgrow** the gains from specialization — and *LRAC* turns upward.

The U-shape of *LRAC* is specialization winning early and coordination losing late.

Compare the **efficient scale** with the **size of the market**:

- Efficient scale **small** relative to demand: many firms fit — a crowded, competitive industry.
- Efficient scale **large** relative to demand: only a few fit — perhaps only **one**.

The shape of *LRAC* is a fundamental determinant of market structure.

Active Learning 6

A parts maker owns the **medium** factory. Per-unit costs at $Q = 2$: small \$4.00, medium \$6.80, large \$23.20.

1. Orders collapse to 2 a day — **permanently**. What is its per-unit cost **right now**? What should it do in the long run, and what does per-unit cost become?
2. Same firm, years later: output grows from 3 to 6; long-run average cost falls from \$2.90 to \$2.00. Name the phenomenon — and give **one mechanism** that produces it.

Say which run you are reasoning in at each step — that distinction is the marks.

Four minutes. Paper and pencil.

PART 7

Pulling It Together

The cost concepts at a glance

Term	Definition	Formula
Explicit costs	require a money outlay	—
Implicit costs	forgone alternatives, no outlay	—
Accounting profit	revenue minus explicit costs	$TR - \text{explicit}$
Economic profit	revenue minus all opportunity costs	$TR - (\text{exp} + \text{imp})$
Fixed cost	does not vary with output	FC
Variable cost	varies with output	$VC(Q)$
Total cost	everything	$TC = FC + VC$
Average fixed cost	fixed cost per unit	$AFC = FC/Q$
Average variable cost	variable cost per unit	$AVC = VC/Q$
Average total cost	cost of the typical unit	$ATC = TC/Q$
Marginal cost	cost of the next unit	$MC = \Delta TC / \Delta Q$

The chain

fixed input



marginal product falls (each worker adds less)



marginal cost rises ($MC = w/MP$)



marginal pulls average (MC crosses ATC at its minimum: efficient scale)



the long run frees the fixed input ($LRAC$: scale economies, then diseconomies)

From costs to supply

The question the cost curves could not yet answer:

How much should the firm produce?

The missing ingredient is the **price** the market pays. The firm will hold price against **marginal cost**, one unit at a time — produce while the unit earns more than it costs.

The cost curves built here are the raw material of the supply curve.

Active Learning 7 — the whole chain

A sign shop: $Q = 20L - 2L^2$ signs a day ($L \leq 5$ workers), wage \$40, fixed bill \$100.

1. Build the table for $L = 0, \dots, 5$: columns Q , TC , and ATC .
2. Marginal cost of a sign for each step: $MC = 40/MP$. What is it doing, and why?
3. Find the **efficient scale**, then **verify it two ways**: (a) compare the ATC column directly; (b) the pulling check — MC below ATC into the bottom, above it out. Say why the two routes must agree.

Five minutes. Paper and pencil. The chain, start to finish, on fresh numbers.

A firm's cost curves are its technology, seen in a mirror.

Marginal product falls, so marginal cost rises.

Marginal below average pulls it down; above, pulls it up —
so where marginal meets average, average is at its minimum.